Whisper STT Implementation - Brief Report

Lab: Whisper STT Implementation
Implementation: Python / Jupyter Notebook
Author: Sergii Rostovskyi

Prompted and unprompted transcription

Actual differences cannot currently be established. When run, both approaches use the same 60-second WAV and identical whisper-1 parameters. The guided approach adds the verified name Project Nimbus. The code saves complete responses and differing excerpts in comparison.json and PNGs. A vocabulary may help with names but can cause incorrect substitutions. A text change does not prove improvement: disputed passages must be checked by listening. Without a reference transcript, no WER or superiority claim is made.

Benefits of chunking

Sequential chunking limits upload sizes and allows retrying only failed excerpts. Responses are cached by the audio hash and parameters. The demonstration uses 60 seconds; for long meetings, 600 can be specified, with an automatic PCM WAV size limit. The total duration is checked. The chunk start time is added to local segment/word timestamps. The tradeoff is more requests and loss of context at boundaries. The accuracy and performance of this strategy have not yet been measured experimentally.

Challenges

The main observed obstacle is the blocking network proxy. The original stt.ipynb uses a microphone, so the input was adapted to the attached MP3. Secure key loading, FFmpeg decoding, size checks, timestamp validation, and JSON/TXT/SRT export are implemented. Code syntax has been checked; a complete run with dependencies and API access is still necessary. The empty example files are explicitly labeled as templates, not transcripts.

Recommendations

Use a clean recording and the correct language, and add only known terms. For long speech, choose boundaries at pauses; when chunks overlap, remove duplicates separately. Check names, numbers, and join points against the audio. Create an independent reference to evaluate WER. After a successful run, replace this preliminary analysis with observations from the comparison and attach the automatically generated PNGs from outputs/png.